Carbohydrate Research 428 (2016) 18–22

Contents lists available at ScienceDirect

Carbohydrate Research

journal homepage: www.elsevier.com/locate/carres

Formation of isomers of anionic hemiesters of sugars and carbonic acid in aqueous medium

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A R T I C L E I N F O

Article history:
Received 29 February 2016
Received in revised form 6 April 2016
Accepted 7 April 2016
Available online 12 April 2016

Keywords:
Hemiester of carbonic acid
Monoalkyl carbonate
Sugar carbonate
Isomers
NMR spectroscopy

A B S T R A C T

Hemiesters of carbonic acid can be freely formed in aqueous media containing HCO_3^-/CO_2 and mono- or poly-hydroxy compounds. Herein, ^{13}C NMR spectroscopy was used to identify isomers formed in aqueous solutions of glycerol (a prototype compound) and seven carbohydrates, as well as to estimate the equilibrium constant of formation (K_{eq}). Although both isomers are formed, glycerol 1-carbonate corresponds to 90% of the product. While fructose and ribose form an indistinct mixture of isomers, the anomers of D-glucopyranose 6-carbonate correspond to 74% of the eight isomers of glucose carbonate that were detected. The values of K_{eq} for the disaccharides sucrose (4.3) and maltose (4.2) are about twice the values for the monosaccharides glucose (2.0) and fructose (2.3). Ribose ($K_{eq} = 0.89$)—the only sugar without a significant concentration of a species containing a $-CH_2OH$ group in an aqueous solution—resulted in the smallest K_{eq}. On the basis of the K_{eq} value and the concentrations of HCO_3^- and glucose in blood, one can anticipate a concentration of 2–4 μmol L^{-1} for glucose 6-carbonate, which corresponds to ca. of 10% of its phosphate counterpart (glucose 6-phosphate).

1. Introduction

About a century ago, Hempel and Seidel[1] and Siegfried and Howwjanz[2] synthesized several hemiesters of carbonic acid (HECAs). They obtained calcium salts of hemiesters of different alcohols and sugars by passing CO_2 into an aqueous solution containing the hydroxy compound and excess of $Ca(OH)_2$. The general formula for these salts is $(ROCO_2)_2Ca$, which can be regarded as one of the carbonic acid hydroxyls that reacted with the alcohols while the other one is left as a free carboxylate.

Sir Haworth extended these studies about sugar carbonates, because he was motivated by the possibility of natural occurrence. He hypothesized the existence of sugar carbonates in plants, which would be formed during the absorption of CO_2 by the leaves.[3,4] He finally stated:[3]

> There can be no doubt that sugars possess the capacity of combining with carbon dioxide, and it appears singular that no evidence is available pointing to the isolation of sugar carbonates from plants.

An important issue in handling of HECA salts is that they easily hydrolyse in water regenerating the alcohol and releasing bicarbonate. The decomposition is an even bigger issue at low pH.[5–8] He realized that this behavior could affect the extraction techniques of sugar derivatives, which could explain the lack of evidence on HECAs in plants.

By the middle of the last century, however, these hypotheses seem to have vanished owing to the persistent lack of proofs about the formation of HECAs in biological medium.[9] Taking into account that (1) the concentration of inorganic carbon (HCO_3^-/CO_2) in living organisms is about 10^{-2} mol L^{-1}, (2) in prevailing aqueous media, the equilibrium

$$ROH + HCO_3^- \rightleftharpoons ROCO_2^- + H_2O \quad (1)$$

is shifted towards the formation of the free alcohol, and (3) the low kinetic barrier for this interconversion, one can understand the nontriviality of an isolation procedure, or even the detection of HECAs in living organisms.

Along the following decades, the HECAs—also known as monoalkyl carbonates (MACs)—were used only as models for other molecules of biochemical relevance[8] and in studies about enzymatic activity.[10,11] However, despite the presence of a great number of hydroxyl compounds and HCO_3^- in almost any compartment of the living cells, no possible role in biochemical processes has been identified for a HECA yet. Actually, this fact is more a consequence of the unawareness about the formation of such species than a proof

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http://dx.doi.org/10.1016/j.carres.2016.04.007
0008-6215/

of the intrinsic futility of HECAs for the metabolism of living organisms. There is a lack of analytical methods for detection of HECAs and fundamental studies about their properties that must be overcome before any attempt to establish biological roles.

In recent years, capillary electrophoresis with capacitively coupled contactless conductivity detection (CE-C^{4}D) allowed the detection of HECAs formed by the reaction of an alcohol and HCO_3^- in aqueous media.[12,13] This technique was used to detect monoethyl carbonate in beers and other carbonated drinks.[14] Subsequently, these findings were confirmed by electrospray ionization-mass spectrometry (ESI-MS),[15] and the limit of detection was improved by using CE–ESI-MS/MS.[16] This advancement in the analytical chemistry for the detection of HECAs was possible not only because of the modern instrumentation, but also because of a careful choice of the experimental conditions in order to preserve these evanescent species.

Not only simple alcohols, but also polyols and sugars have been investigated by CE, MS, and CE–MS. However, questions about the formation of isomers still remain. Taking into account a typical biological medium and the equilibrium constants, one can expect that only mono substituted species would be detected at significant levels.[13] Therefore, due to the different reactivity of the hydroxyls of a sugar, different amounts of the isomers should be formed. Herein, we report the detection and identification of isomers of anionic sugar carbonates by using ^{13}C NMR, which is, to the best of our knowledge, achieved for the first time.

2. Results and discussion

The presence of the carbonate group induces changes in the chemical shift of the backbone carbon atoms of the alcohol as demonstrated by Sauers et al.[8] At the same time, changes to the chemical shift of the carbonate group (when compared to free HCO_3^-) are observed as shown by Richardson et al.[17] However, due to the ^{13}C NMR low sensitivity, these peaks are not easily detected if the medium is primarily aqueous (see Eq. (1)). To access HECAs in aqueous medium by ^{13}C NMR, relatively high concentrations of $H^{13}CO_3^-$ and the alcohol as well as long acquisition time are needed. Therefore, the study was carried out using 0.2 mol L^{-1} $NaH^{13}CO_3$ solutions, and the spectra were accumulated for a long time (over a weekend) to improve the signal-to-noise ratio (SNR).

2.1. Glycerol carbonate

Glycerol was initially studied as a prototype for the sugars, because of the presence of only two distinct hydroxyls (primary and secondary). Fig. 1 shows the ^{13}C NMR spectra, in the vicinity of the bicarbonate peak, of solutions containing glycerol, $NaH^{13}CO_3$, and a mixture of them in D_2O. As previously observed,[13] the reaction between bicarbonate and glycerol takes place in a matter of minutes. Therefore, the spectra, which were obtained by accumulation during ca. 60 h, are representative of the dynamic equilibrium state.

The spectrum of the mixture shows two new peaks (162.18 and 161.73 ppm), which were attributed to the isomers of glycerol carbonate and clearly observed at upfield of the HCO_3^- peak (163.19 ppm). Although usually the intensities of the peaks for ^{13}C NMR cannot be directly related to the abundance, this approximation was acceptable in the present case, because of the similarity of the chemical environment and proximity of the chemical shifts. Therefore, one can estimate the equilibrium constant of formation (K_{eq}) according to Eq. 1 as follow. Starting with a bicarbonate concentration b and no HECA, when the equilibrium is reached, the HECA concentration is x, and bicarbonate concentration is b–x. Taking into account the proportionality of the peak area and concentration, the HECA concentration is now given by

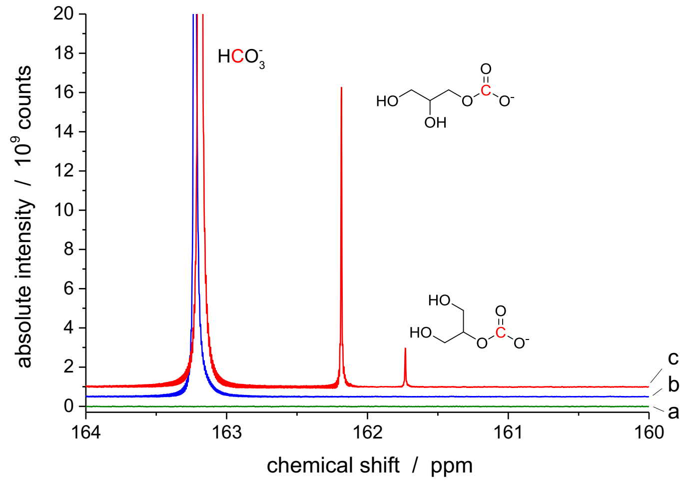

Fig. 1. ^{13}C NMR spectra of aqueous solutions of glycerol plus NaCl (a), $NaH^{13}CO_3$ (b), and glycerol plus $NaH^{13}CO_3$ (c).

$$x = \frac{A_H}{A_B + A_H} b \tag{2}$$

where A_H and A_B are the peak areas for the HECA and bicarbonate, respectively. Finally, the equilibrium constant of formation is given by

$$K_{eq} = \frac{x(w+x)}{(a-x)(b-x)} \tag{3}$$

where w and a are the initial concentrations of water and alcohol, respectively.

Taking the sum of the peak areas of the glycerol carbonate isomers and the one for HCO_3^-, the equilibrium constant of formation (K_{eq}) of glycerol carbonate was estimated at 2.1, which is in good agreement with 2.2 previously obtained by CE-C^{4}D.[13] The individual K_{eq} values for the isomers were 1.9 and 0.2, which show that one of them greatly prevails.

The simulation of the ^{13}C NMR spectra (ChemBioDraw v. 14, CambridgeSoft) indicated that the carbon atoms at positions 1 and 2 in the glycerol structure are shifted by +5.9 and −3.6 ppm for glycerol 1-carbonate and −2.5 and +3.6 ppm for glycerol 2-carbonate. Small peaks shifted by +4.0 and −1.9 ppm were detected in the spectrum of the mixture of glycerol and $NaHCO_3$ (Fig. S1), which suggests that glycerol 1-carbonate is the most abundant isomer. Unfortunately, contrary to the glycerol carbonate, the identification of the isomers based on the $\Delta\delta$ caused by the carbonate group on the backbone carbon atoms could not be successfully applied to the other sugars, because of the complexity of the spectra and the low SNR of the backbone peaks of the anionic sugar carbonates.

The tendency to form greater amount of the HECA at the primary carbon atom is in agreement with a previous observation about the correlation between K_{eq} and pK_a of the hydroxyl group.[8,13] On the basis of estimated pK_a values and experimental K_{eq} values, Sauers et al[8] estimated that log(K_{eq}) varies according to 1.4 times the pK_a value of the hydroxy group of a monohydric alcohol. Although this finding could be used to predict the relative amount of HECAs in a mixture of alcohols, it cannot be straightly applied to a polyol, because the tendency to a new ionization diminishes after the ionization of a first hydroxyl. Therefore, the pK_a to be considered in the present case would be that one for the ionization of a specific hydroxyl keeping neutral all the other ones. This value has been obtained by simulation (Chemicalize, ChemAxon, Budapest, Hungary) of a polyol changing all hydroxyls but one by a methoxyl, which prevents successive ionizations to take place.[13]

The estimated values of pK_a for the primary and secondary glycerol hydroxyls, using the suggested approach, are 14.6 and 13.7, respectively, which translate into a $\Delta\log(K_{eq})/\Delta pK_a$ equals to 1.0. Taking into account all the experimental differences, this value approaches the 1.4 predicted by the Sauers' model. This result suggests that the pK_a values of the polyol hydroxyls may be used as a guide to predict the relative abundances of the corresponding carbonate isomers.

It is worth noting that, even after 60 h of data accumulation, the spectra for glycerol and all the sugars do not show peaks at the region of 156–154 ppm. The simulation of the ^{13}C NMR spectra (ChemBioDraw v. 14, CambridgeSoft) and previous experimental results[18] show that this is the region for full carbonate esters. Therefore, although hemiesters were always detected, there is no evidence of the formation of full esters of the sugars and carbonic acid.

2.2. Glucose carbonate

Similarly to the previous case, the ^{13}C NMR spectrum of pure D-glucose showed no peak from 160 to 164 ppm. The same is true for two other derivatives: methyl α-D-glucopyranoside (G1M) and 3-O-methyl-D-glucopyranose (G3M). However, the ^{13}C NMR spectra of mixtures with bicarbonate showed the formation of HECA isomers for all of them (Fig. 2).

Fig. 2b shows the spectrum of a solution of glucose plus $NaH^{13}CO_3$, in which eight isomers were detected. The sum of the peak areas of all these isomers leads in a global K_{eq} of 2.0, which is significantly greater than the previous value of 0.35 obtained by CE-C^{4}D.[13]

The CE-C^{4}D method is based on the migration of the anionic HECAs and detection at two different points a few minutes after they are removed from the sample plug. Due to the similarity of the mobilities, the isomers of the glucose carbonate migrate as one peak and, thus, only a global K_{eq} value can be obtained. However, this approach is reliable only if all the isomers have the same hydrolysis kinetics, which is not necessarily true. Therefore, the K_{eq} equals to 2.0, which was obtained by ^{13}C NMR of a solution at dynamic equilibrium, is a more reliable value. This result also suggests that the isomers have different kinetic of hydrolysis, which led to the previous underestimated K_{eq} value of 0.35, because part of the isomers would be completely or partially hydrolysed at the detection points in the electrophoresis experiments.

In aqueous solution, glucose is found mainly as α-pyranose (38%) and β-pyranose (62%),[19] and each one has five hydroxyl groups performing a total of 10 possible isomers. The detection of only eight isomers suggests that one of the hydroxyl groups does not yield the corresponding HECA at a significant level. If one considers that (1) the α- and β-anomers have similar constants of formation and (2) the positional isomerism is the most relevant effect, the eight peaks could be arranged in pairs with the ratio 38/62 for their areas—the same proportion of the reacting anomers. In Fig. 2b, the best attempt to identify these pairs keeping simultaneously the ratios as close as possible to the expected value is shown. Although no structure can be attributed to this point, the peaks of the anomeric forms of four isomers (**x**, **y**, **w**, and **z**) were, thus, identified.

Fig. 2c shows the spectrum of G1M and $NaH^{13}CO_3$. As expected, only four isomers were detected, because isomerization was precluded by the methyl group at position 1 (anomeric carbon), which keeps the glucose skeleton in its α-anomeric form. Despite the small upfield shift, there was a good agreement between the chemical shifts and the relative intensities of the four peaks for G1M and α-glucose. Taking into account that the carbonate at position 1 of G1M is not allowed to be formed, and that the four isomers are similar for both carbohydrates, one can conclude that glucose 1-carbonate is the missing isomer.

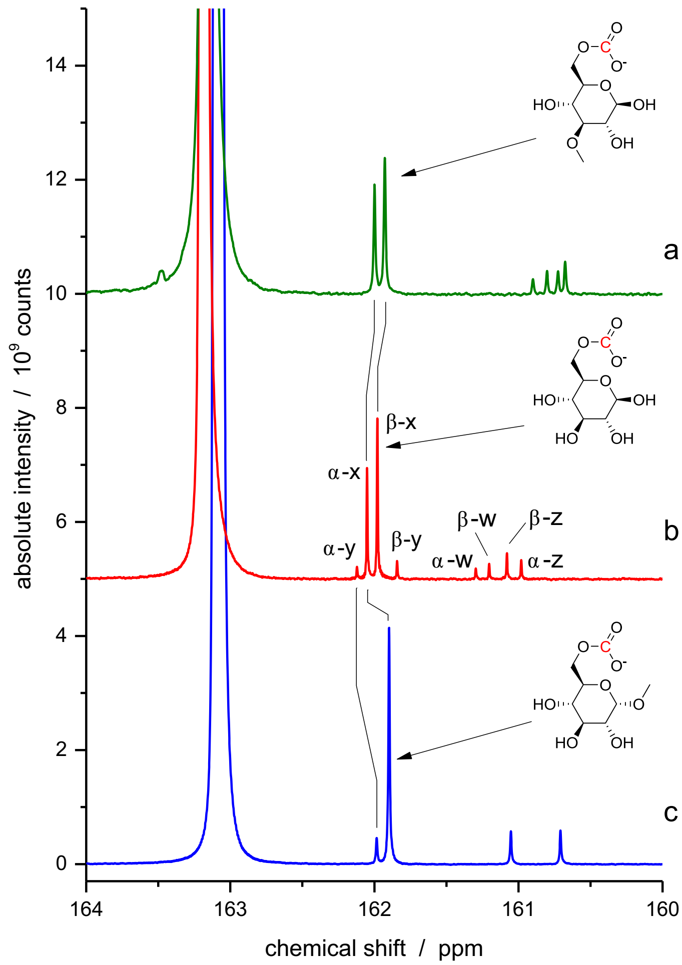

Fig. 2. ^{13}C NMR spectra of aqueous solutions containing $NaH^{13}CO_3$ and G3M (a), glucose (b), or G1M (c). The α–β pairs for glucose carbonate were attributed keeping simultaneously the peak area ratios as close as possible to 38/62. G1M renders only the α-isomers, which can be associated to the equivalent α-isomers of glucose carbonate. According to this attribution, glucose 1-carbonate is the non-detected isomer. The relative intensities of the peaks and the pK_a values of the hydroxyl groups suggest that glucose 6-carbonate (β-d-glucopyranose 6-carbonate and α-d-glucopyranose 6-carbonate) is the most abundant isomer (x). The spectrum of G3M carbonate suggests that the isomer y is glucose 3-carbonate. The structures are the most abundant anomeric form of the most abundant positional isomer for each case.

The pK_a of the hydroxyl at position 1 is ca. 11.3—the smallest among the five hydroxyls of glucose (Fig. S2). Actually, HECAs of alcohols with such a low pK_a value used to be also not detected by the previous techniques.[13,15] These results suggest that, although a priori glucose 1-carbonate can be formed, its concentration is much smaller than the concentrations of the other isomers.

The peak areas from Fig. 2b suggest that isomer **x** is dominant, and that together the α- and β-forms account for 74% of the detectable glucose carbonate. One more time, the empirical correlation between K_{eq} and pK_a of the hydroxy group allows us to suppose that glucose 6-carbonate is the most abundant isomer and that the other three isomers have similar abundances (Fig. S2). Therefore, although it is not conclusive, the hypothesis of being glucose 6-carbonate the dominant isomer **x** is reinforced.

Although the hydroxyls at position 2, 3, and 4 have different pK_a values (12.4, 13.0, and 13.1, respectively), this information was not used for a tentative identification of the peaks for the remaining isomers, because small differences in pK_a values could lead to misinterpretations. However, additional information was obtained by using glucose with a methoxy group at position 3 (G3M). Fig. 2a

Table 1
Equilibrium constant of formation and main isomer of HECAs

Polyol	K_{eq}[a] (^{13}C NMR)	K_{eq}[b] (CE–C^{4}D)	Main isomers[c]
Glycerol	2.1	2.2	Glycerol 1-carbonate (90%)
D-Glucose	2.1	0.35	D-glucopyranose 6-carbonate (74%)
G1M	1.7	–	Methyl α-D-glucopyranoside 6-carbonate (70%)
G3M	1.6	–	3-O-methyl-D-glucopyranose 6-carbonate (76%)
D-Fructose	2.3	–	D-fructopyranose carbonate[d]
D-Ribose	0.89	–	D-ribose carbonate[d]
Sucrose	4.3	1.1	Sucrose 6-carbonate + sucrose 6′-carbonate (65%)
Maltose	4.2	–	Maltose 6′-carbonate (47%) Maltose 6-carbonate (35%)

[a] Polyol concentration ≤ 1 mol L^{-1}, D_2O, 25 °C, pH 8.1, I = 0.2 mol L^{-1}.
[b] Results from do Lago et al.[13] for H_2O, 20 °C, pH 8.3, I = 0.03 mol L^{-1}.
[c] Tentative attribution based on ^{13}C NMR spectra and pK_a values of the hydroxyl groups.
[d] Mixture of isomers not identified.

shows the spectrum of G3M and NaH^{13}CO$_3$. In this case, the methoxy group is not attached to the anomeric carbon, which allows the α- and β-forms to be observed at equilibrium. The absence of peaks related to the **y** isomer in this spectrum indicates that this isomer is glucose 3-carbonate. Consequently, the isomers **w** and **z** are the isomers with carbonate at positions 2 and 4.

2.3. *Other anionic hemiesters of sugars and carbonic acid*

Fig. 3 shows the spectra of four other sugars: maltose, sucrose, ribose, and fructose. By comparison with glucose carbonate, the main peaks in the maltose carbonate spectrum (Fig. 3a) were tentatively attributed to the isomers esterified at position 6 of each glucose group. Maltose anomers are in equilibrium in aqueous solutions. Therefore, one should expect a similar effect to that one observed in the glucose carbonate spectrum. However, the effect tends to be smaller for the positions far from the anomeric centre, which should lead to smaller differences in the chemical shift of the anomeric forms. This criterion was used to do the final attribution shown in Fig. 3a.

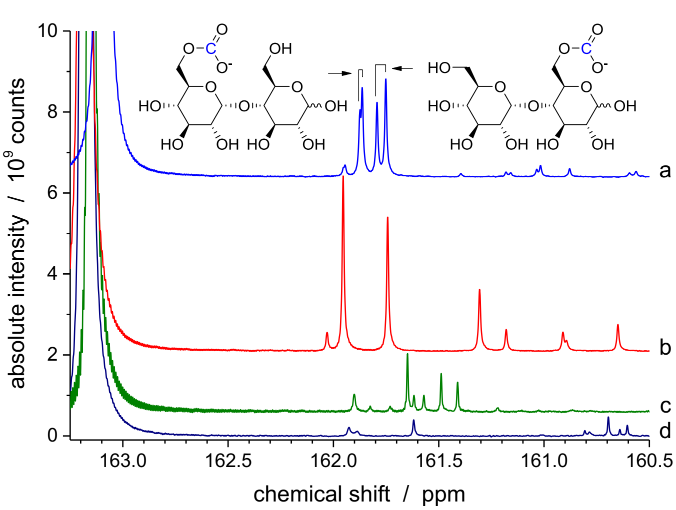

Fig. 3. ^{13}C NMR spectra of aqueous solutions of NaH^{13}CO$_3$ and maltose (a), sucrose (b), fructose (c), and ribose (d). Maltose 6-carbonate and sucrose 6-carbonate were attributed by similarity with glucose 6-carbonate. Maltose 6′-carbonate is the only second option for a primary alcohol in the whole structure. The peak at 161.74 ppm could not be unequivocally attributed to either sucrose 1′-carbonate or sucrose 6′-carbonate.

According to the pK$_a$ values for sucrose (Figure S3), the hydroxyls at position 6 in both glucose and fructose residues are equivalent. Therefore, the two most intense peaks in sucrose carbonate spectrum (Fig. 3b) were attributed to the isomers sucrose 6-carbonate and sucrose 6′-carbonate, which stands for 65% of the products.

Fructose and ribose are examples of carbohydrates that exist in a great variety of forms in aqueous equilibrium. Therefore, although they have different abundances, a bunch of isomers of anionic sugar carbonates were detected for each one. On the basis of abundances of the fructose cyclic forms in aqueous medium, d-fructopyranose carbonates should be the most abundant isomers. Ribose is an even more complicated case, because although the pyranose form is most abundant (79%),[20] only the furanose form has a primary alcohol group with the highest pK$_a$ value (Figure S4). Therefore, an unequivocally attribution cannot be done. Nevertheless, a global equilibrium constant can still be calculated.

Table 1 summarizes the findings for all the sugars that were studied. Similar to glucose carbonate, sucrose carbonate seems to have isomers with different kinetic of hydrolysis, resulting in an underestimate value of K_{eq} by CE-C^{4}D. Fructose carbonate was also previously detected by CE-C^{4}D,[13] but the fast hydrolysis has precluded any estimation of its K$_{eq}$ at that time. The present results suggested that glucose and fructose carbonates have similar K_{eq} values but different kinetic of hydrolysis. Sucrose and maltose (disaccharides) carbonates have K_{eq} values about twice the monosaccharide ones, which seem to be related with the total concentration of hydroxyl groups in the solution. Ribose, the only sugar without a significant concentration of an isomer containing a primary alcohol group, has the smallest K_{eq} value for the formation of its hemiester, which reinforces the previous observation that primary alcohols tend to form the correspondent hemiester in a greater extent.

3. Conclusions

To the best of our knowledge, this is the first time that the formation of HECA isomers of polyols in aqueous medium has been demonstrated. In some cases, such as for fructose and ribose, an indistinct mixture of isomers is formed. However, there is a clear trend of specific isomer formation for glucose.

For the sake of illustration, let us think over the presence of glucose carbonate in blood. On the basis of the obtained value of

K_{eq} for glucose 6-carbonate (K_{eq} = 1.5 for D-glucopyranose 6-carbonate) and the concentrations of glucose and CO_2/HCO_3^-, one can anticipate that the glucose 6-carbonate concentration in blood is 2–4 µmol L^{-1}, while its phosphate counterpart (glucose 6-phosphate) is ca. 30 µmol L^{-1}. Although this concentration of the carbonate ester is about 10 times smaller than the phosphate ester, both of them are anionic forms of glucose having similar structures. Moreover, taking into account the low kinetic barrier for the formation of glucose 6-carbonate and that the stock of inorganic carbon is about ten times greater than the stock of inorganic phosphate, there would be a remarkable ability to replenish the glucose 6-carbonate stock, if it would be consumed in some biological process. Therefore, the possible involvement of glucose 6-carbonate in metabolic pathways as well as in glucose transportation should be investigated. In a broader sense, these results allow us to redeem the former Haworth hypothesis about the presence of anionic sugar carbonates in plants and expand it to any living organism.

4. Experimental procedure

The following solutions were prepared using D_2O with 0.75% (w/w) sodium 3-(trimethylsilyl)propionate-2,2,3,3-d_4 and 0.2 mol L^{-1} $NaH^{13}CO_3$: 1 mol L^{-1} glycerol, 0.5 mol L^{-1} d-glucose, 1.0 mol L^{-1} methyl α-d-glucopyranoside (G1M), 0.5 mol L^{-1} 3-O-methyl-d-glucopyranose (G3M), 0.5 mol L^{-1} d-fructose, 0.5 mol L^{-1} d-ribose, 0.5 mol L^{-1} sucrose, 0.5 mol L^{-1} maltose. The same set of solutions was also prepared using NaCl instead of $NaH^{13}CO_3$ for reference purposes. A 0.2 mol L^{-1} $NaH^{13}CO_3$ solution containing none of the polyols was also used as reference. The 125.8 MHz 1D 13C-[1H] spectra of these 17 solutions were acquired at 25 °C using pulse sequence with 1H decoupling on a Bruker Avance III 500 spectrometer (Rheinstetten, Germany) equipped with a 5 mm TXI Probe. Typical conditions for the ^{13}C spectra were 81,920 transients, spectral width of 32.9 kHz, and 64k data points giving an acquisition time of 0.99 s and FID resolution of 0.50 Hz/point.

Acknowledgements

This work was supported by FAPESP (grant 2012/06642-1). C.L.L. would like to thank CNPq (researcher fellowship 304415/2013-8). D.T.R.V. and V.B.S. would like to thank FAPESP (fellowships 2011/02156-2 and 2013/14993-1).

Supplementary material

Supplementary data to this article can be found online at doi:10.1016/j.carres.2016.04.007.

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